Multi-Arm Bandit Problems

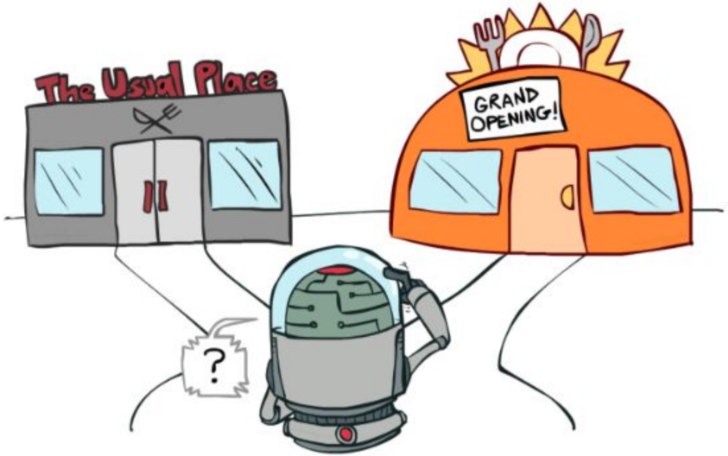
DECISION THEORY

The theory behind balancing the **Explore** vs. **Exploit** dilemma for optimization.

The Saturday Night Dilemma

THE SCENARIO

It's Saturday night and you're hungry. How do you decide where to go?



EXPLORE

Try new places and discover something amazing.

EXPLOIT

Stick with your favorite. Safe, but limited.

THE MULTI-ARMED BANDIT PROBLEM

Decision depends on: current option quality vs. alternative uncertainty.

The History

HISTORICAL BACKGROUND

In the 1950s, statisticians began studying the math behind gambling machines.

KEY CHALLENGE

The "Multi-Armed Bandit" problem: choosing between a known reward and a potentially better unknown.

THE CHALLENGE

Statisticians faced a critical question: How do you maximize winnings when you don't know the payout probability of each machine?

Bandits in the Wild

APPLICATIONS

- A/B Testing
- News Article Ranking
- Ad Placement
- Recommendations
- Dynamic Pricing
- Website Optimization

Optimizing Decisions

In the real world, the "arms" are options presented to users. Multi-Arm Bandit algorithms allow systems to maximize immediate rewards while simultaneously gathering data to improve future performance.

Three Strategies

1. ALWAYS EXPLORE

Try machines randomly

- Learn everything about the environment.
- **Outcome: You lose money!**

2. ALWAYS EXPLOIT

Stick with the first winner.

- Conservative strategy.
- **Outcome: Miss the best machine.**

3. BANDIT ALGORITHMS

The best of both worlds.

- Explore sometimes.
- Exploit most of the time.
- **Outcome: Optimal balance.**

The Goal

Maximize total reward over time OR minimize regret.

REGRET FORMULA

Regret = Expected Optimal Reward - Reward Received

How much reward did we miss out on?

Epsilon Greedy: Decision Logic

A simple strategy for balancing
discovery and reward.

THE ALGORITHM

With probability ε → EXPLORE

Otherwise → EXPLOIT

CORE INTUITION

Occasionally try something new. By introducing controlled randomness, we prevent being trapped by "good enough" and keep searching for "the best".

Upper Confidence Bound (UCB)

Prefer options that are **uncertain**.

THE FORMULA

$$\text{UCB1}_i(t) = \hat{\mu}_i + \sqrt{\frac{2\ln t}{n_i}}$$

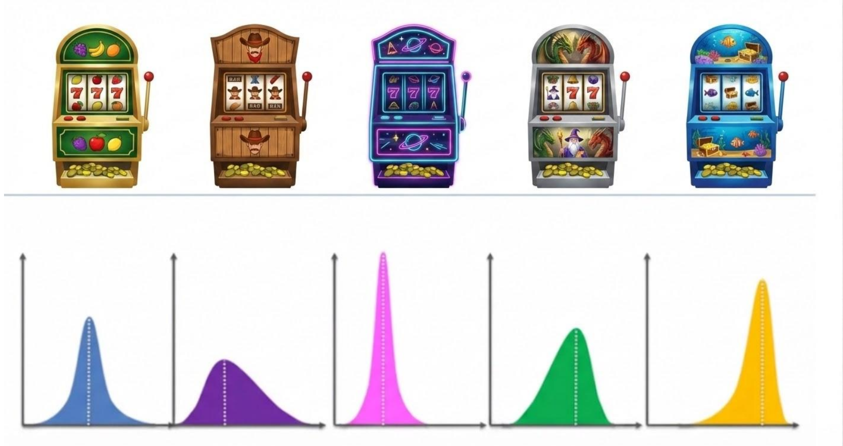
score = average reward + uncertainty bonus

CORE INTUITION

Algorithm tracks **average reward** and **uncertainty**. Options with fewer trials get a larger bonus. Forces exploration while ensuring quick convergence.

Thompson Sampling (Probabilistic)

Choose according to the probability that an arm is **best**.



THE STRATEGY

1. Model belief about each arm.
2. Sample from that belief.
3. Pick the best sample.

CORE INTUITION

Introduces **probabilistic randomness**, which avoids overcommitting too early. It balances exploration by naturally favoring high-potential outcomes.

A/B TESTING: WHY BANDITS BEAT NAIVE TESTING

TRADITIONAL A/B TEST

Traffic split: A = 50%, B = 50%, even if B is clearly better.

THE PROBLEM:
You are wasting half your traffic on sub-optimal options.

BANDIT APPROACH

Traffic evolves over time (e.g., Day 1: 50/50; Day 5: 10/90).

ADVANTAGES:

- Learn faster
- Fewer users see bad options
- Continuous optimization

SUMMARY

A/B testing → **Learn first, earn later.**

Bandits → **Learn while earning.**

News Article Ranking: Avoiding Stagnation

THE SETUP

- 5 slots / 30 possible articles
- Goal: Maximize clicks (Reward: 1 or 0)
- Problem: Popularity bias kills discovery

BANDIT SOLUTION

Algorithm balances showing hits with giving **new exposure** to fresh content.

Why It Matters

Without bandits, the homepage **stagnates**. With them, new content surfaces quickly, driving higher long-term engagement.

All Sorts of Crazy Bandits!



Bandit Type	Idea / Intuition	Use Case
Contextual	Use context/features to pick arms	Personalization
Adversarial	Robust to fluctuating rewards	Ad placement
Sliding Window	Weight recent rewards more	Changing worlds
Bayesian UCB	Pick arm using upper quantile	Optimism
Pure Explore	Identify best arm quickly	Clinical trials
Gaussian/Cont.	Model rewards as smooth function	Continuous params

But wait...

While the variety is vast, the core remains: balancing **Exploration** and **Exploitation** to find the best possible outcome in an uncertain environment.

Mastering the Uncertain

Life is a multi-armed bandit problem

TAKEAWAYS

- Don't exploit too early; you'll miss the best arm.
- Don't explore too long; you'll waste your reward.
- Bandit algorithms give us the math to guide our intuition

Make Out Like a Bandit

Try to earn while you learn to optimize your results and decrease your **FOMO.**